Elias Hadjipanayis

3274

Dr. Boudreau

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Homework 10

Chapter 11 Exercise 5: Spinning Top.

Integrate the equation of the spinning top. Plot the three variables u, ψ, and ϕ vs. time.

Solution:

We start with the Lagrangian of the system in Euler's angles.

$$\mathrm{L} = \left(\frac{I_1}{2}\right)\left(\dot{\theta}^2 + \dot{\varphi}^2\sin^2\theta\right) + \left(\frac{I_3}{2}\right)\left(\dot{\psi} + \dot{\varphi}\cos\theta\right)^2 - \mathrm{Mglcos}\,\theta$$

Where I_1 is the moment of inertia about the transverse axes and I_3 is the moment of inertia about the symmetric axis. Using substitutions,

$$Let\ u = \cos(\theta) \rightarrow \dot{u} = \sin(\theta)\,\dot{\theta}$$

$$\dot{\theta}^2 = \frac{\dot{u}^2}{(1-u^2)}$$

$$sin^2(\theta) = 1 - u^2$$

$$\mathrm{L} = \left(\frac{I_1}{2}\right)\left[\frac{\dot{u}^2}{(1-u^2)} + \dot{\varphi}^2(1-\mathrm{u}^2)\right] + \left(\frac{I_3}{2}\right)(\dot{\psi} + \mathrm{u}\dot{\varphi})^2 - \mathrm{Mglu}$$

Then, I obtained the conjugate momenta and inverted to get the velocities in terms of them. In this case, we can use H = T + V since the kinetic energies are quadratic and the Lagrangian is time independent. So:

$$H = \frac{\left((1-u^2)p_u^2\right)}{(2I_1)} + \frac{\left(p_\varphi - up_\psi\right)^2}{\left(2I_1(1-u^2)\right)} + \frac{\left(p_\psi^2\right)}{(2I_3)} + Mglu$$

With the velocities.

$$\dot{u} = \frac{(1-u^2)p_u}{I_1}$$

$$\dot{\varphi} = \frac{p_\varphi - up_\psi}{I_1(1-u^2)}$$

$$\dot{\psi} = \frac{p_\psi}{I_3} - \frac{u\left(p_\varphi - up_\psi\right)}{I_1(1-u^2)}$$

I then used the PhaseSpace object from the Classical namespace to contain all of my variables and their initial values. I used the RungeKuttaSolver with my Hamiltonian and phase space and solved each ODE.

Chapter 11 Exercise 9: Separable Hamiltonians.

Write and test a symplectic integrator that will integrate any separable Hamiltonian, along the lines of Section 11.5.2. Use the automatic derivative system of QatGenericFunctions.

Solution:

A separable Hamiltonian splits into a kinetic part that depends only on the momenta and a potential part that depends only on the coordinates:

$$H(q,p) = T(p) + V(q)$$

Because of this, $\partial H/\partial q_i$ depends only on the coordinates and $\partial H/\partial p_i$ depends only on the momenta, so the momenta and coordinates can be updated separately. I used the leapfrog method, where each step size h is a half kick, a full drift, and another half kick:

$$p_i \leftarrow p_i - \frac{h}{2}\frac{\partial H}{\partial q_i}$$

$$q_i \leftarrow q_i + h\frac{\partial H}{\partial p_i}$$

$$p_i \leftarrow p_i - \frac{h}{2}\frac{\partial H}{\partial q_i}$$

I wrote a SymplecticIntegrator class that takes the Hamiltonian as a Genfun function, a PhaseSpace, and the step size h.

To test it, I used the harmonic oscillator

$$H = \frac{1}{2}(p^2 + q^2)$$

starting at q = 1, p = 0, and integrated to t = 10000 with h = 0.3 using both my integrator and QAT's RungeKuttaSolver with a fixed-step classical RK4 stepper at the same step size. I then plotted the energy error $E - E_0$ vs. time for both. The RK4 energy decays steadily, while the leapfrog energy error stays bounded between $-h^2/8$ and 0 for the entire run. I am not completely sure why this occurs.

Chapter 11 Exercise 13: The SEIR Disease Model.

The SEIR disease model divides a population into susceptible, exposed, infectious, and recovered groups, with births at rate b and deaths in each group at rate d. (a) Write the four rate equations for the schematic, with force of infection $\lambda = \beta I/N$. (b) If b = d, is the population static? (c) What does the model assume about disease mortality? (d) Solve the equations with $\beta = 0.2$, $\gamma = 0.1$, $b = d = 10^{-4}$, $\alpha = 0.1$, $\nu = 10^{-3}$, $I_0 = 10^{-6}$, N = 1, and $S_0 = N - I_0$, and plot the infected population ratio on a log scale for several thousand days.

Solution:

a) Following the schematic, people are born into S at rate b, move from S to E at rate λ, from E to I at rate α, from I to R at rate γ, and lose their immunity from R back to S at rate ν. Every group dies at rate d. This gives:

$$\frac{dS}{dt} = bN - \lambda S - dS + \nu R$$

$$\frac{dE}{dt} = \lambda S - \alpha E - dE$$

$$\frac{dI}{dt} = \alpha E - \gamma I - dI$$

$$\frac{dR}{dt} = \gamma I - \nu R - dR$$

where the force of infection is

$$\lambda = \beta \frac{I}{N}, \qquad N = S + E + I + R$$

b) Adding all four equations,

$$\frac{dN}{dt} = (b - d)N$$

So if b = d, the population is static. In my program, N = 1.

c) The model assumes the disease is never fatal. Infected people die at the same background rate d as everyone else, which is also why the total population does not change.

d) I wrote the four rate equations as Genfun functions of the Variables S, E, I, and R and gave them to QAT's RKIntegrator using addDiffEquation, along with their initial values. I then used getFunction to obtain S(t), E(t), I(t), and R(t). Then, I plotted I/N on a log scale from 0 to 5000 days. The infected ratio rises to about $2 * 10^{-2}$ near day 400, then crashes to about $5 * 10^{-5}$ near day 800 because most of the susceptible population has been infected. Once people start losing their immunity, a second, smaller outbreak peaks near day 1400. Each outbreak is weaker than the last, and I/N settles toward a constant 5.5×10^{-3} by the end of the run.